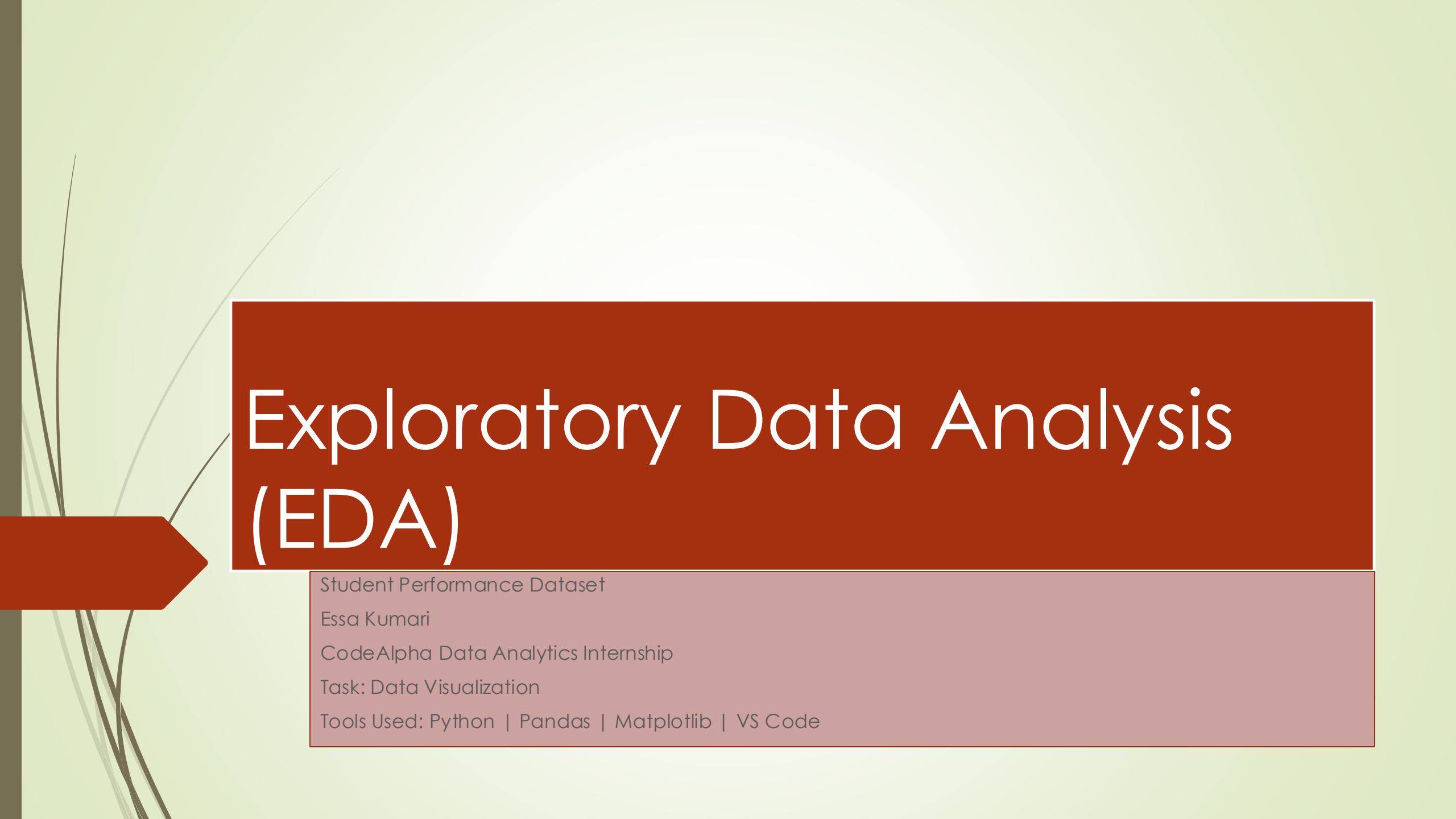




Exploratory Data Analysis (EDA)

Student Performance Dataset

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Task: Data Visualization

Tools Used: Python | Pandas | Matplotlib | VS Code



Introduction

- Exploratory Data Analysis (EDA) is the process of understanding and examining a dataset before performing detailed analysis. It helps identify patterns, trends, and relationships in the data through statistics and visualizations.
 - In this project, I analyzed a Student Performance Dataset containing students' gender, mathematics, reading, and writing scores. Using Python and Matplotlib, I created different charts to better understand the dataset and draw meaningful insights.
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Project Objectives

- 
- Load the student dataset using Python.
 - Understand the structure of the dataset.
 - Compare student performance in different subjects.
 - Create meaningful visualizations.
 - Draw useful insights from the data.



Dataset Description

Column	Description
Gender	Gender of the Student
Math Score	Mathematics Marks
Reading Score	Reading Marks
Writing Score	Writing Marks

Body

Total Records: 10

Total Columns: 4



Tools and Technologies Used

- Python 3.13
 - Pandas
 - Matplotlib
 - Visual Studio Code (VS Code)
- 



Loading the Dataset

- The dataset was loaded using the Pandas library.
- `import pandas as pd`
- `data = pd.read_csv("dataset/student.csv")`
- After loading the dataset, I verified that the file was imported successfully before starting the analysis.



Data Preview

- The first five rows of the dataset were displayed using:
- `print(data.head())`
- This helped confirm that the dataset was loaded correctly and that all columns contained the expected values.
- *(Insert the screenshot of data.head() on this slide.)*

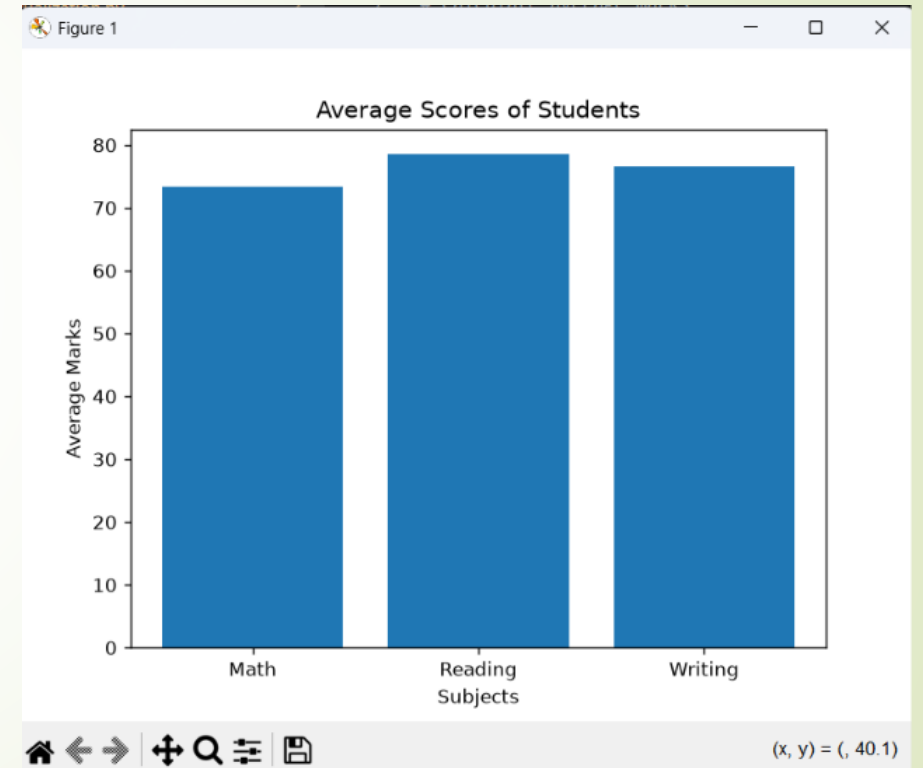
Bar Chart Analysis

➤ Purpose

- To compare the average scores of Mathematics, Reading, and Writing.

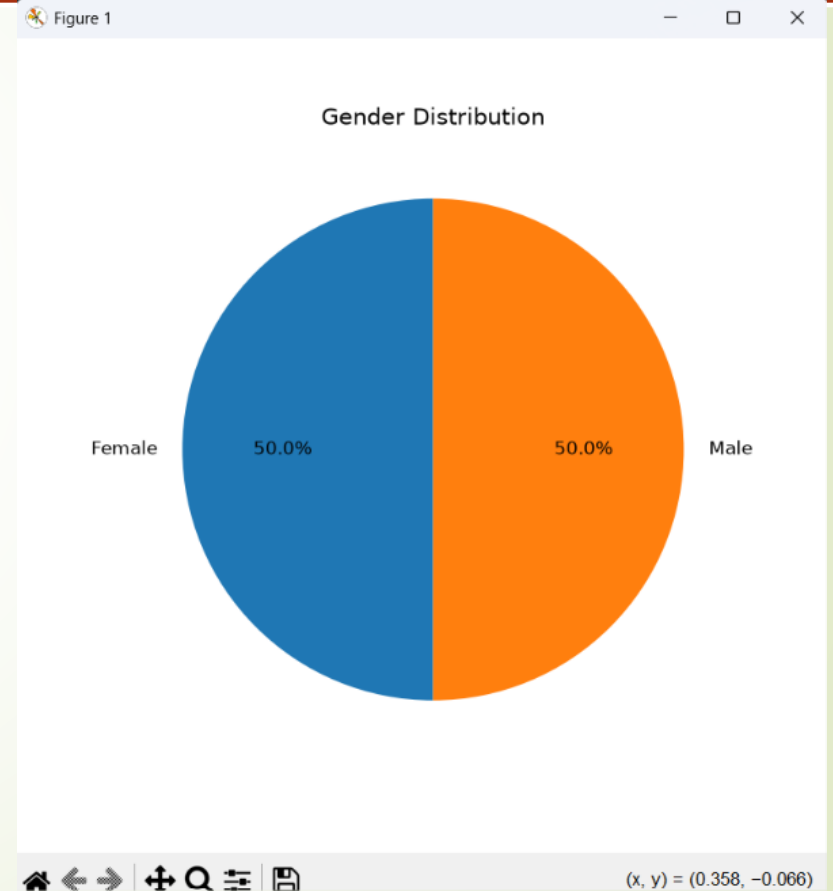
➤ Observation

- The average Reading and Writing scores are slightly higher than the Mathematics score.



Pie Chart Analysis

- **Purpose**
- To show the percentage of Male and Female students.
- **Observation**
- The dataset contains an equal number of male and female students.



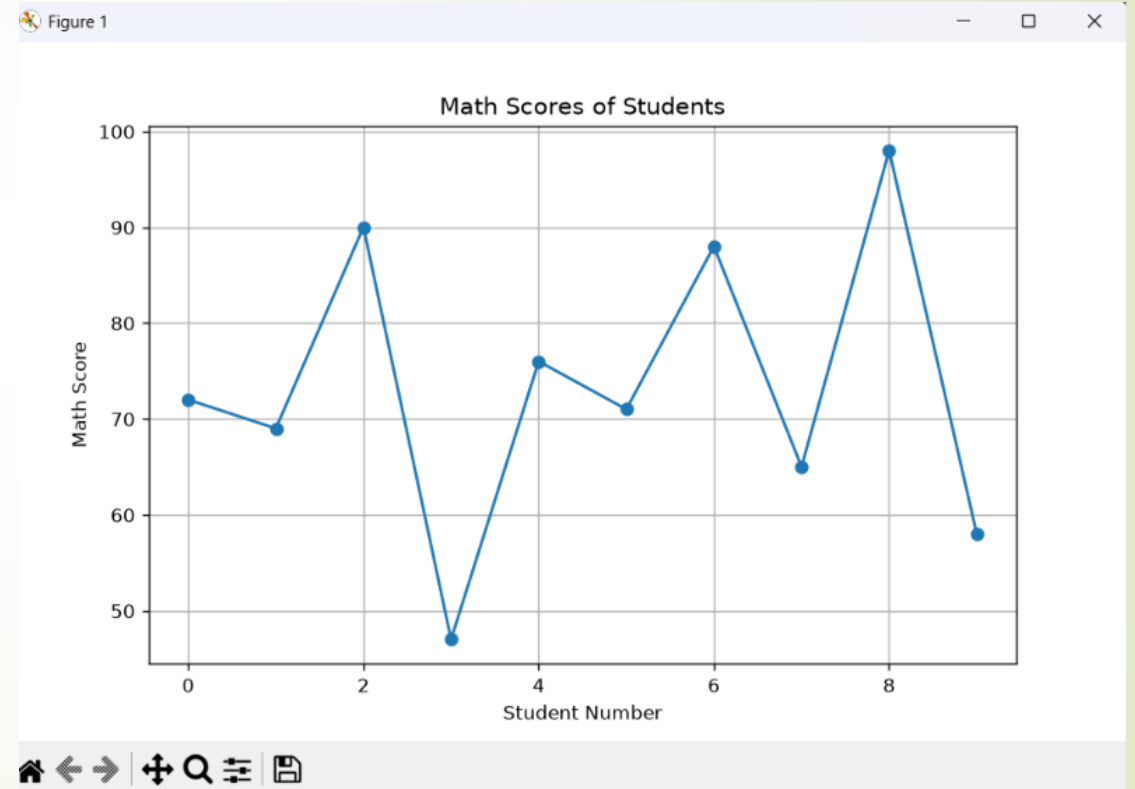
Line Chart Analysis

➤ Purpose

- To display Mathematics scores of all students.

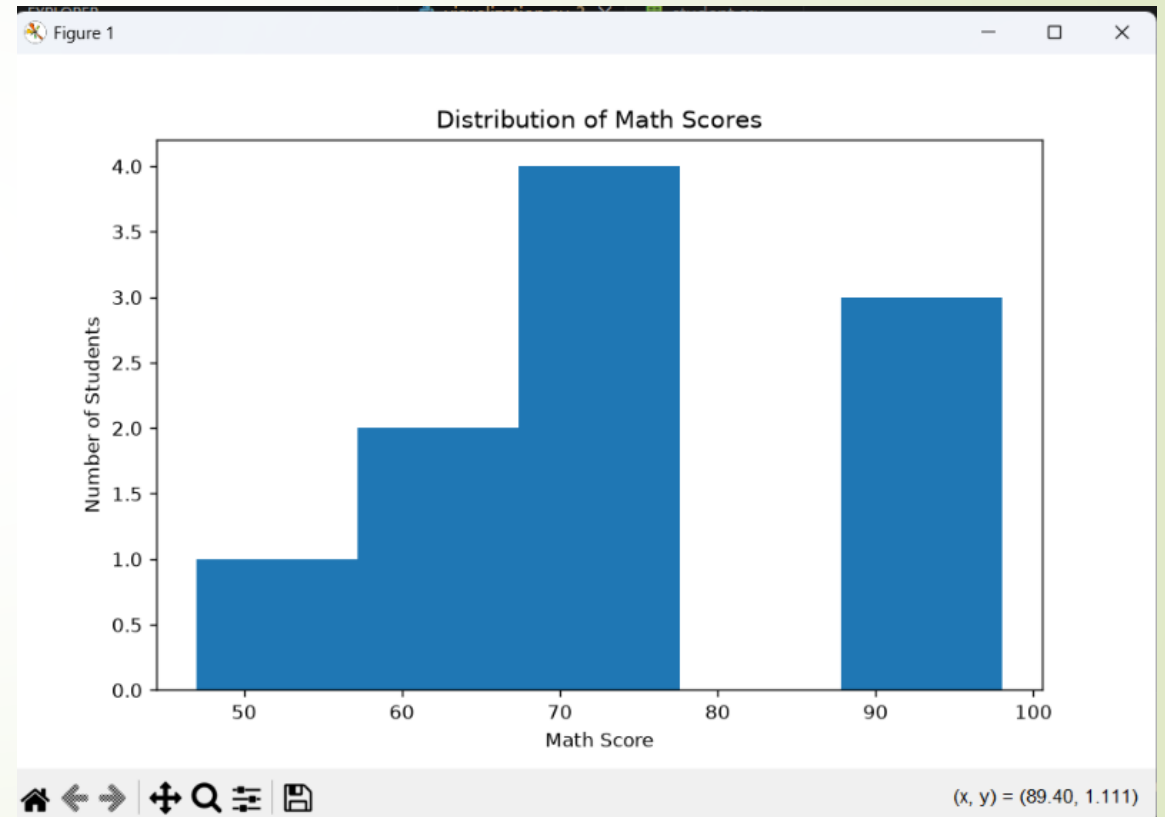
➤ Observation

- The scores vary from one student to another, showing differences in performance.



Histogram Analysis

- **Purpose**
- To understand the distribution of Mathematics scores.
- **Observation**
- Most students scored between **60 and 90 marks**.





Key Findings

- Equal representation of male and female students.
 - Reading and Writing scores are slightly higher than Mathematics.
 - Most students performed well.
 - No missing values were found in the dataset.
 - Visualizations made the data easier to understand.
- 



Conclusion

- This project helped me understand how Exploratory Data Analysis (EDA) is performed using Python.
- By creating different visualizations, I was able to study student performance and identify useful patterns in the dataset.
- This task also improved my understanding of data visualization and strengthened my Python skills.



Future Scope

- 
- Analyze larger educational datasets.
 - Apply Machine Learning for prediction.
 - Create interactive dashboards using Power BI or Tableau.
 - Compare performance across different schools.
 - Perform advanced statistical analysis.



Thank You

Submitted by

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